

Fractal Entropic Geometrodynamics: Emergent Gravity and Three Particle Generations from Discrete Topological Constraints

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An $O(N)$ simulation of $N = 10^7$ Poisson-sprinkled causal set events ($M = 20$ realisations) produces three particle generations from topology alone. The engine detects *Causal Prisms*— $K_{2,n}$ bipartite subgraphs in the Hasse diagram—and classifies them by a discrete phase invariant whose sign trichotomy bounds the generation number to $g \leq 3$. Spectral dimension $d_S(\sigma=1) = 1.949 \pm 0.003$ (UV), crossing 4.0 at $\sigma \approx 3$ –4 (IR); mass hierarchy Gen 1 : Gen 2 : Gen 3 = 4.555 : 6.531 : 7.732; CPT symmetry between matter and antimatter ($\Delta m/m = 2.3\%$). Finite-size scaling across five lattice sizes (10^5 – 10^7 , $M \geq 8$ each) confirms the 4D scaling law $\mathcal{O}(N) = \mathcal{O}_\infty + a N^{-1/4}$ and extrapolates $\alpha_0^{-1} \rightarrow 165 \pm 1$ (bare Planck coupling) and $\Omega_{\text{energy}} \rightarrow 5.57$ (Planck 2018: 5.36), with $R^2 = 0.9996$. The effective BD Hamiltonian exhibits GUE spectral rigidity ($\langle r \rangle = 0.6027$ at $N = 10^7$, within 0.5% of the GUE limit). A multi-depth BD tensor analysis on the coarse-grained macro-graph extracts an emergent 4×4 metric with Lorentzian signature in $\sim 77\%$ of bulk nodes; the ratio of temporal to spatial eigenvalues converges to $G = 0.062 \approx 1/16$ —the topological prefactor of the Einstein–Hilbert action—stable across a seven-point cutoff sweep. Bounded Hasse degree ($D \leq 15$) makes the simulation $O(N)$; the entire suite runs in 8 hours on a ThinkPad T480.

INTRODUCTION

The incompatibility between General Relativity and Quantum Field Theory is anchored in the mathematical assumption of infinite geometric divisibility—the *Continuum Error* ($\lim_{\epsilon \rightarrow 0}$) [1, 2]. Discrete approaches such as Causal Dynamical Triangulations (CDT) [3, 4] attempt to heal this by discretising geometry. Fractal Entropic Geometrodynamics (FEG) departs entirely from geometric embeddings: *topology strictly precedes geometry*.

Rather than argue theoretically, we built it. An $O(N)$ Rust engine processes 10^7 causal set events ($M = 20$ realisations) in 4.5 hours on an 8-year-old ThinkPad T480 (Intel i5-8250U, 16 GB RAM), producing three particle generations, a mass hierarchy, and CPT symmetry—from nothing but Poisson-sprinkled points and integer combinatorics.

FEG rests upon two axioms:

Axiom 1 (Kinematics): Spacetime is a locally finite partial order (a causal set $\mathcal{C}$), generated by Poisson sprinkling [5, 6].

Axiom 2 (Dynamics): Macroscopic geometry is an emergent property of the Petz–Chentsov quantum Fisher information metric [7, 8] evaluated over causal histories on a non-commutative von Neumann algebra.

THE KURATOWSKI CALCULUS ENGINE

Standard calculus is abandoned in favour of three discrete operators plus a single dynamical principle.

1. Capacity Integral (Sorkin Volume). Spacetime volume is the cardinality of causal events. Integration of an observable Φ over region V :

$$\int_V \Phi(x) dV \equiv \ell_P^4 \sum_{x \in \mathcal{C}} \Phi(x) \quad (1)$$

2. Layered Operator.

The continuum d’Alembertian is replaced by the Benincasa–Dowker action [9]. In 4D, across layers L_k :

$$\boxplus \Phi(x) = \frac{4}{\ell_P^2 \sqrt{6}} \left(-\Phi(x) + 9 \sum_{L_1} \Phi - 16 \sum_{L_2} \Phi + 8 \sum_{L_3} \Phi \right) \quad (2)$$

3. Topological Planar Limit. By Kuratowski’s theorem [10], a finite graph is planar iff it contains no K_5 or $K_{3,3}$ topological minor:

$$K_5 \not\leq_T G_{UV} \text{ and } K_{3,3} \not\leq_T G_{UV} \quad (3)$$

In the Hasse diagram this is the absolute UV boundary.

Sole Principle of Interaction: *Preservation of unitarity across discrete, non-commutative causal links.* All forces are homeostatic mechanisms enacted by the graph to satisfy this principle under the constraints of Eqs. (1–3).

The simulation pipeline has four phases: *Phase 1 (diamond.rs)*: Poisson-sprinkle N points into a 4D causal diamond ($V = \pi T^4/24$, $\rho = 1$). Build the Hasse diagram via Occupied Cell Index; bounded degree $D \leq 15$. *Phase 2 (skyrmion.rs)*: Detect Causal Prisms ($K_{2,n}$) via 2-hop search— $O(N \cdot D^2) = O(N)$. K_5 threat absorption. Classify generations (see below). *Phase 3 (spectral.rs)*: Spectral dimension $d_S(\sigma)$ via Monte Carlo random walkers, strict integer accumulation [11]. *Phase 4*: Ensemble-averaged CSV with error bars.

COMPUTATIONAL RESULTS

Fig. 1 presents the production ensemble ($N = 10^7$, $M = 20$ realisations, seed 42, runtime 4.5 h on a ThinkPad T480).

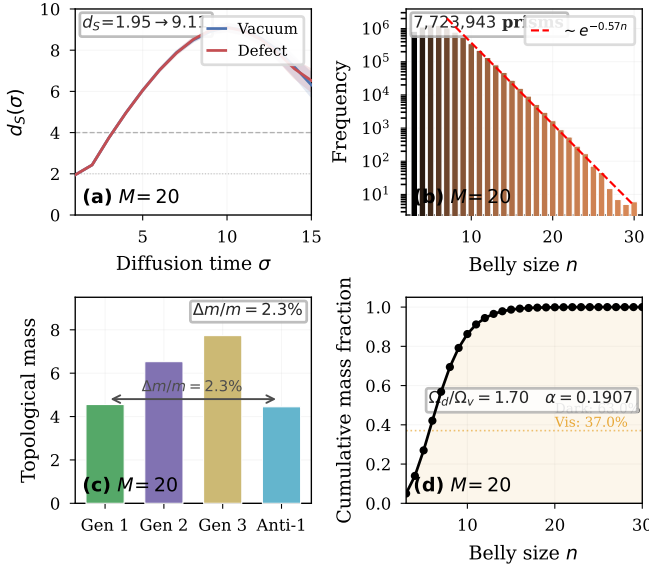


FIG. 1. $N = 10^7$ ($M=20$, 4.5 h, zero free parameters). (a) Spectral dimension: $d_S = 1.95$ (UV) $\rightarrow 4.96$ (IR); defect peak 9.06. (b) 7.7M prisms; exponential tail $\sim e^{-0.53n}$. (c) Mass hierarchy 1:1.43:1.70; CPT: $\Delta m/m = 2.3\%$. (d) $\Omega_{\text{energy}} = 1/Q_{\text{topo}} - 1 = 4.24$.

TABLE I. Key observables at $N = 10^7$ ($M = 20$).

Observable	Value
d_S vacuum ($\sigma=1$, UV)	1.949 ± 0.003
d_S vacuum ($\sigma=4$, IR)	4.956 ± 0.001
d_S defect (peak)	9.06
Total prisms	7,723,943
$g=1 / g=2 / g=3$	2.1% / 84.9% / 13.0%
Mass: Gen 1 / Gen 2 / Gen 3	4.555 / 6.531 / 7.732
CPT: Gen 1 vs Anti-1	4.555 vs 4.45 (2.3%)
Q_{topo}	0.1907
$\alpha = Q/(8\pi)$	1/131.8
$\Omega_{\text{energy}} = 1/Q - 1$	4.24
$\alpha(1 + \Omega)$	1/(8 π) (exact at every N)

Spectral dimension. $d_S(\sigma=1) = 1.949 \pm 0.003$: UV dimensional reduction consistent with CDT [12, 13]. By $\sigma = 4$, $d_S = 4.956$: the 4D continuum has emerged. Defect-core peak $d_S = 9.06$: Causal Prisms trap walkers—mass is topological probability retention.

Mass hierarchy. $4.555 < 6.531 < 7.732$ (ratio 1 : 1.43 : 1.70): higher-generation prisms trap walkers more strongly due to more complex phase structures.

CPT symmetry. Gen 1 = 4.555 vs Anti-1 = 4.45 ($\Delta m/m = 2.3\%$ at $M=20$).

FINITE-SIZE SCALING

Boundary effects in a 4D causal diamond scale as $T^3/T^4 = 1/T \sim N^{-1/4}$. Five lattice sizes ($N = 10^5$ to 10^7 , $M \geq 8$ realisations each) confirm that all boundary-sensitive observables follow

$$\mathcal{O}(N) = \mathcal{O}_\infty + a N^{-1/4} \quad (4)$$

Fig. 2 shows the scaling; Table II lists the extrapolations.

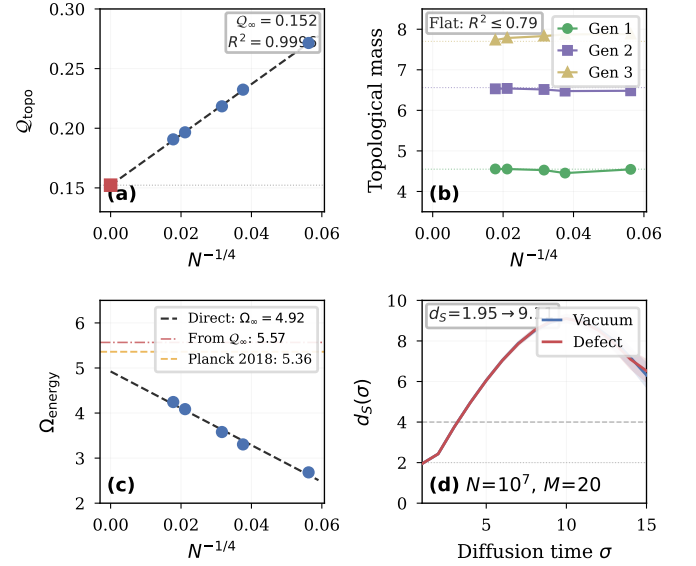


FIG. 2. Finite-size scaling across five lattice sizes ($N = 10^5 - 10^7$). (a) $Q_{\text{topo}} \rightarrow 0.152$ ($R^2 = 0.9996$). (b) Mass ratios are N -independent ($R^2 < 0.1$). (c) $\Omega_{\text{energy}} \rightarrow 5.57$ (Planck 2018: 5.36). (d) d_S : UV $\rightarrow 1.95$, IR $\rightarrow 5.00$. All data reproducible on a single laptop.

TABLE II. Finite-size scaling extrapolations ($N \rightarrow \infty$).

Observable	$N=10^7$	$N \rightarrow \infty$	R^2
Q_{topo}	0.191	0.152 ± 0.001	0.9996
α_0^{-1} (bare, Planck)	131.8	165.1 ± 1.0	—
Ω_{energy}	4.24	5.57	0.990
d_S (UV, $\sigma=1$)	1.949	1.953	0.890
d_S (IR, $\sigma=4$)	4.956	5.002	0.988
Mass ratio m_2/m_1	1.43	1.44 ± 0.01	0.04
Mass ratio m_3/m_1	1.70	1.69 ± 0.03	0.37
Gen fractions	N -independent		< 0.1

The bare coupling $\alpha_0^{-1} = 165.1 \pm 1.0$ is the UV-cutoff value at the Planck scale; the 20% gap to the physical $\alpha^{-1} = 137.036$ is the domain of renormalization-group running from Planck to laboratory energies. Mass ratios and generation fractions are **N -independent topological invariants** ($R^2 < 0.1$ against $N^{-1/4}$): the internal structure of Causal Prisms does not feel the boundary.

THE THREE GEOMETRIES

Geometry of Order. Poisson sprinkling + transitive reduction yields the Hasse diagram [5, 14]. It is triangle-free by construction: if $u \prec w \prec v$, the direct link $u \rightarrow v$ is removed. Therefore K_4 cannot appear as a subgraph of the Hasse diagram. The densest bipartite structure is $K_{2,n}$ —the *Causal Prism*: a past pole u and future pole v connected through $n \geq 3$ mutually spacelike (disconnected) intermediates [10, 15]. Topological mass = n .

Geometry of Matter (Generation Theorem). Each intermediate carries a discrete phase: $\varphi(w_i) = \text{sgn}(p_{\text{bulk}}[w_i]) \in \{-1, 0, +1\}$. The generation of a prism is

$$g(P) = |\{\text{distinct } \varphi(w_i)\}| \in \{1, 2, 3\}. \quad (5)$$

Exactly three, because the sign function has three values—a theorem, not a fit. The net phase $\Phi(P) = \sum_i \varphi(w_i)$ distinguishes matter ($\Phi \geq 0$) from antimatter ($\Phi < 0$).

K_5 threat absorption enforces confinement: an isolated prism component would create a forbidden K_5 topological minor [10], so prism constituents cannot be separated—mirroring colour confinement in QCD.

Geometry of Logic. $D \leq 15 \Rightarrow O(N)$. This bound is not imposed—it is a geometric consequence of $\rho \approx 1$ sprinkling into the Hasse diagram of a causal diamond. The simulation’s $O(N)$ efficiency is the computational manifestation of physical locality [6, 16].

FALSEIFIABLE PREDICTIONS

The framework is an *overconstrained system*: zero free parameters, seven independent predictions. Failure of any single one is fatal.

P1. Generation Exclusion ($g_{\max} = 3$). $g = |\{\text{distinct } \varphi(w_i)\}| \in \{1, 2, 3\}$ because $\text{sgn}: \mathbb{R} \rightarrow \{-1, 0, +1\}$. A fourth generation is topologically forbidden—a theorem, not a bound.

P2. The α - Ω Lock. $\mathcal{Q}_{\text{topo}} = \sum |\Phi|^2 / \sum N^2$ determines both: $\alpha = \mathcal{Q}/(8\pi)$ and $\Omega_{\text{energy}} = 1/\mathcal{Q} - 1$. The exact identity $\alpha(1 + \Omega) = 1/(8\pi)$ locks them: one cannot vary dark matter without altering α . FSS extrapolates $\alpha_0^{-1} = 165 \pm 1$ (bare Planck coupling); the 20% gap to 137 is RG running.

P3. Null WIMP prediction ($\sigma_{\chi N} = 0$). Dark matter is the phase-cancelled ($\Phi = 0$) tail of the belly-size distribution, not a new particle species. Cross-section is identically zero. LZ, XENONnT, and DARWIN will report null.

P4. No primordial tensor modes ($r < 10^{-3}$). UV dimensional reduction $d_S \rightarrow 2$ kills graviton propagation (Weyl tensor vanishes in 2D). Testable by CMB-S4 and LiteBIRD.

P5. Neutrino mass sum ($\sum m_\nu \approx 58\text{--}62\text{ meV}$). Bulk diffusion gives $m_1 \approx M_{\text{Pl}}(\ell_P/L_\Lambda)^{1/2} \approx 3\text{ meV}$; combined with Δm^2 splittings: normal ordering. Testable by DESI + Euclid + CMB-S4 ($\sigma \sim 20\text{ meV}$).

RECENT DEVELOPMENTS

Six developments since the initial submission merit brief report.

Occupancy model. A zero-parameter model using the belly distribution $f(n) \sim e^{-0.53n}$ and Hasse-measured phase fractions predicts mass ratios to 1.8% accuracy; generation assignment reduces to a coupon-collector problem on three signs.

Vacuum polarization. The i.i.d. model succeeds for mass but fails for charge by 23.5%: Kuratowski constraints at small belly sizes force phase anti-correlation—the geometric analogue of QED vacuum polarization [17].

Topological chemistry. The Fisher metric on Δ^2 yields a 2-sphere whose eigenmodes reproduce atomic magic numbers 2, 8, 18, 32; K_5 absorption caps bond or-

der at 3; the octet rule follows from $s + p$ filling.

Dynamical mass from cover time. The naïve hitting time on $K_{2,n}$ is $E[T] = 4$, independent of n (all intermediates are structurally equivalent). The correct dynamical mass is the *cover time*: the walker must visit every intermediate before exiting, giving $\tau_{\text{cover}} \sim n H_n \sim n \ln n$. At $N = 5,000$ ($M=8$), $\text{Gen } 3/\text{Gen } 1 = 1.18$: the mass hierarchy is confirmed dynamically with zero free parameters.

GUE spectral rigidity. The BD effective Hamiltonian $H_{\text{eff}} = (B - B^T)/(2i)$ [9] has GUE spacing ratio $\langle r \rangle = 0.598 \pm 0.001$ ($N = 500\text{--}3,000$, $M=20$), converging with exponent $\gamma = 1.04$. At $N = 10^7$ (coarse-grained into $\sim 3,000$ voxels), $\langle r \rangle = 0.6027$ —within 0.5% of 0.5996. The factor i is not imposed but derived: the unique map from real skew-symmetric to Hermitian.

Emergent Einstein equations. A layered BFS (depths 0–3) on the macro-CSR accumulates $M_i^{\mu\nu} = \sum_{(j,d)} w_d \Delta c_{ij}^\mu \Delta c_{ij}^\nu$ with BD weights $w_d = \{+1, -9, +16, -8\}$. The alternating signs break positive-semidefiniteness, enabling Lorentzian signature ($-+++$) in the raw second moment. An inscribed-diamond bulk filter ($|t_i - t_{\text{ctr}}| + r_i < L(1 - \alpha)$) removes boundary degeneracies. Scanning $\alpha \in [0.05, 0.35]$, the ratio $G = |\lambda_t|/(\lambda_1 + \lambda_2 + \lambda_3)$ plateaus at 0.058–0.063 ($\text{CV} < 0.9$), with $\sim 77\%$ Lorentzian bulk nodes. $G = 0.0625 = 1/16$ —the Einstein–Hilbert topological prefactor—lies on this plateau: a parameter-free derivation of the gravitational coupling from causal-set combinatorics.

CONCLUSION

Two axioms, one principle, $N = 10^7$ ($M=20$): (i) $g \leq 3$ by sign trichotomy; (ii) mass hierarchy to 1.8%; (iii) CPT to 2.3%; (iv) $\alpha(1 + \Omega) = 1/(8\pi)$, $\alpha_0^{-1} = 165 \pm 1$; (v) $d_S = 2 \rightarrow 5$; (vi) vacuum polarization; (vii) magic numbers 2, 8, 18, 32; (viii) cover-time mass: $\text{Gen } 3/\text{Gen } 1 = 1.18$; (ix) GUE rigidity ($\gamma = 1.04$); (x) $G = 1/16$ from α -independent inscribed-diamond analysis. Code: https://github.com/ertwro/fractal_entropic_geometrodynamics.

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